6.1 Properties of Exponents

Name _____ Period ____ Date _____

Algebra I 6.1 (30-minute plan) · Student sheet

I can…

- use zero and negative exponents.
- use the properties of exponents to simplify expressions.
- use the properties of exponents with scientific notation.

Vocabulary · fill in the missing word

Zero exponent — a^0 = _____

Negative exponent — a^{-n} = _____

Product of powers — $a^m \cdot a^n$ = _____

Quotient of powers — a^m/a^n = _____

Power of a power — $(a^m)^n$ = _____

Power of a product / quotient — $(ab)^m$ = _____ / $\left(\frac{a}{b}\right)^m = a^m/b^m$

Part 1 · Notes and examples (fill in each note, then work the example)

Zero and negative exponents: a^0 = _____ a^{-n} = _____ $(a \neq 0)$

Example 1

Evaluate each expression.

a. 6.7^0 b. $(-3)^{-4}$ c. $\frac{5^0}{2^{-2}}$

Product, quotient, and power of a power: $a^m \cdot a^n$ = _____ a^m/a^n = _____ $(a^m)^n$ = _____

Simplified form: An expression is simplified when it has no _____ exponents and each base appears _____.

Example 2

Simplify. Write your answer using only positive exponents.

a. $x^3 \cdot x^{-5}$ b. $\frac{w^{10}}{w^4}$ c. $(z^4)^{-3}$

Power of a product and of a quotient: $(ab)^m$ = _____ $(a/b)^m$ = _____

Example 3

Simplify.

a. $(-1.5y)^2$ b. $\left(\frac{a}{-10}\right)^3$ c. $\left(\frac{3d}{2}\right)^4$

Part 2 · Partner practice

Partner practice

Simplify. Write your answer using only positive exponents.

$(2x^3y^{-2})^{-3}$